

HYDROGENATED NBR (HNBR) SYNTHESIS, DEGRADATION, AND FORMULATION TRENDS ACROSS SELECTED PATENTS

1. ABSTRACT

The selected patent set does not describe a single unified HNBR polymerization route; instead, it spans precursor NBR synthesis, post-polymerization molecular-weight reduction by metathesis, hydrogenation to HNBR, and downstream compounding of HNBR in functional articles. Across the evidence, the target material is consistently defined as hydrogenated nitrile rubber in which the C=C bonds of the butadiene-derived units are partially or fully hydrogenated, typically with residual unsaturation in the low single-digit to mid-teens percent range depending on grade and process. The most technically relevant patents are those addressing low-Mooney or low-molecular-weight HNBR production by metathesis degradation of NBR followed by hydrogenation, especially EP1894946B1 and EP2473279B1, which emphasize catalyst/additive control to improve degradation efficiency, lower molecular weight, and narrow polydispersity. EP1894946B1 specifically highlights phosphane or diphosphane additives with Grubbs-type catalysts, while the broader background in EP2473279B1 summarizes prior art routes using co-olefins, ruthenium/osmium/molybdenum/tungsten catalysts, and subsequent hydrogenation to obtain HNBR with Mw roughly 30,000–250,000 and PDI below about 2.5. Other patents in the manifest are formulation-focused rather than synthesis-focused: US10844663B2 uses HNBR as a curable elastomer in fiber-reinforced stator liners; EP1921511B1 uses HNBR in foamed toner supply rollers with zinc methacrylate and peroxide/foaming systems; US20180022951A1 uses HNBR or NBR in electroconductive silver paste, emphasizing low alkali metal content and emulsion-polymerized feedstocks; US9656446B2 includes HNBR among multilayer elastomeric barrier articles; and US11106161B2 mentions HNBR as one possible elastomeric layer material in printing-related intermediate transfer members. Overall, the landscape shows a historical progression from conventional hydrogenation of NBR to more process-efficient metathesis-assisted molecular-weight tailoring, followed by extensive application-specific compounding to balance processability, mechanical durability, conductivity, or barrier performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2057205B1
- Patent Title: Production of meta-block copolymers by polymer segment interchange
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1.: - Example Extracted Example Example 2.: - Example Extracted Example Example 3.: - Example Extracted Example Example 3.: - Example Extracted Example Example 4.: - Example Extracted Example Example 4.: - Example Extracted Example Example 5.: - Example Extracted Example Example 6.: - Example Extracted Example Example 6.: - Example Extracted Example Example 7.: - Example Extracted Example Example 8.: - Example Extracted Example Example 9.: - Example Extracted Example Example 10.: - Example Extracted Example Example 10.: - Example Extracted Example Example 1: - Example Extracted Example Example 1: - Example Extracted Example Example 1: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 2:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 2

- Patent Number: EP1894946B1
- Patent Title: Method for metathesis decomposition of nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 3: - Example Extracted Example example 1.:

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Patent 3

- Patent Number: EP2473279B1
- Patent Title: Metathesis of a nitrile rubber in the presence of a transition metal catalyst
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example comparative Example 45: - Example Extracted Example comparative example 56.: - Example Extracted Example comparative Example 56.: - Example Extracted Example example a: - Example Extracted Example example 1.: - Example Extracted Example example 30: - Example Extracted Example example 40: - Example Extracted Example example 0.: - Example Extracted Example example 12: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4: - Example Extracted Example Example 5: - Example Extracted Example Example 6: - Example Extracted Example EXAMPLE 7.: - Example Extracted Example Example 7: - Example Extracted Example Example 7: - Example Extracted Example Example 7: - Example Extracted Example Example 7:

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Patent 4

- Patent Number: US11106161B2
- Patent Title: Intermediate transfer members for use with indirect printing systems and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Example 1.: - Example Extracted Example Example 2.: - Example Extracted Example Example 3.: - Example Extracted Example Example 4.: - Example Extracted Example Example 5.: - Example Extracted Example Example 4: - Example Extracted Example Example 6.: - Example Extracted Example Example 7.: - Example Extracted Example Example 8.: - Example Extracted Example Example 9.: - Example

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Patent 5

- Patent Number: US10837947B2
- Patent Title: Modified resins and uses thereof
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 6

- Patent Number: US10844663B2
- Patent Title: Fiber reinforced elastomeric stator
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - No specific experimental examples disclosed.

Technical Relevance Selected via deterministic pipeline scoring.

Patent 7

- Patent Number: US9656446B2
- Patent Title: Articles and methods of making the same
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 8

- Patent Number: US20180022951A1
- Patent Title: Electroconductive silver paste
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Example Comparative Example 1: - Example Extracted Example Comparative Example 2: - Example Extracted Example Comparative Example 3: - Example Extracted Example Comparative Example 4: - Example Extracted Example Comparative Example 5: - Example Extracted Example Preparation Example 1: - Example Extracted Example Preparation Example 2: - Example Extracted Example Preparation Example 1:

- Example Extracted Example Preparation Example 3: - Example Extracted Example Preparation Example 1: - Example Extracted Example Preparation Example 4: - Example Extracted Example Preparation Example 1: - Example Extracted Example Example 1: - Example Extracted Example Example 2: - Example Extracted Example Example 3: - Example Extracted Example Example 4: - Example Extracted Example Example 5: - Example Extracted Example Example 6: - Example Extracted Example Example 7: - Example Extracted Example Example 8:

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Patent 9

- Patent Number: EP1921511B1
- Patent Title: Toner supply roller
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - No specific experimental examples disclosed.

Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP1894946B1 and EP2473279B1 are the only patents in this set that directly address HNBR precursor modification by metathesis degradation; both rely on ruthenium-based metathesis chemistry, but EP1894946B1 explicitly adds phosphane/diphosphane co-additives to the catalyst system, whereas EP2473279B1 emphasizes broader transition-metal catalyst selection and prior-art comparisons.
- EP1894946B1 reports metathesis degradation of nitrile rubber followed by hydrogenation to HNBR, with the stated objective of lowering molecular weight and narrowing PDI; the evidence indicates catalyst/additive ratios on the order of 0.01–1 equivalent phosphane or diphosphane per equivalent catalyst, and examples showing improved degradation rate and lower Mw/Mn relative to phosphane-free controls.
- EP2473279B1 discloses HNBR as an end product of NBR metathesis/hydrogenation and summarizes prior art ranges rather than a single optimized recipe; the background cites HNBR with Mw about 30,000–250,000, Mooney viscosity 3–50, and PDI <2.5 from

metathesis-plus-hydrogenation routes, including Grubbs II systems that can operate without co-olefin in some cases.

- US10844663B2, EP1921511B1, US20180022951A1, US9656446B2, and US11106161B2 are formulation/application patents rather than synthesis patents; they use HNBR as a curable elastomer or rubber component and therefore do not provide comparable monomer ratios, polymerization temperatures, or hydrogenation conditions for HNBR manufacture.
- US20180022951A1 is notable for specifying HNBR as a nitrile-group-containing rubber with low alkali metal content (4,000 ppm or less, preferably 1,500 ppm or less) and for indicating that the precursor rubber is preferably emulsion-polymerized and washed to remove salts; this is a feedstock-quality trend rather than a hydrogenation recipe.
- EP1921511B1 and US10844663B2 both show HNBR used in highly filled, crosslinked elastomer systems; EP1921511B1 emphasizes zinc methacrylate loading, foaming agents, and peroxide/foaming-assistant systems for toner rollers, while US10844663B2 emphasizes aramid fibers, silica, and carbon black for stator liners.
- US9656446B2 and US11106161B2 include HNBR only as one option among many elastomers in multilayer or adhesive/release-layer constructions; they reinforce the trend that HNBR is selected for durability and chemical resistance rather than being synthesized in situ within those patents.
- Across the set, the recurring technical theme is that HNBR performance is strongly governed by precursor NBR molecular weight, residual unsaturation, and impurity control; newer approaches focus on lowering Mw/PDI before hydrogenation, while downstream patents focus on compounding to preserve processability and mechanical integrity.

4. CONCLUSION

The evidence supports a clear synthesis strategy for HNBR: prepare a nitrile rubber precursor with controlled unsaturation and low ionic contamination, reduce molecular weight by metathesis when low-Mooney HNBR is desired, then hydrogenate to the target residual unsaturation. Among the patents reviewed, EP1894946B1 provides the most specific process improvement for HNBR precursor degradation, namely the use of phosphane or diphosphane additives with a metathesis catalyst to accelerate degradation and reduce molecular weight while maintaining gel-free product. EP2473279B1 corroborates that metathesis/hydrogenation routes can deliver HNBR with Mw in the tens to hundreds of thousands and PDI below about 2.5. For practical synthesis, the best-supported parameters are: ruthenium-based metathesis catalysts, optional phosphane/diphosphane additive at roughly 0.05–0.5 equivalents per catalyst for efficient degradation, followed by hydrogenation to achieve residual diene unsaturation in the low single-digit to low-teens percent range. For precursor quality, low alkali metal content and emulsion-polymerized NBR feedstocks are favored. Downstream, HNBR is repeatedly selected for high-durability, chemically resistant, and processable elastomer formulations, especially where low Mooney viscosity, controlled crosslinking, or reinforcement are required.

5. REFERENCES

- EP2057205B1 | Production of meta-block copolymers by polymer segment interchange | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2057205B1/en>
- EP1894946B1 | Method for metathesis decomposition of nitrile rubbers | Not disclosed | EP | 2011 | <https://patents.google.com/patent/EP1894946B1/en>
- EP2473279B1 | Metathesis of a nitrile rubber in the presence of a transition metal catalyst | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473279B1/en>
- US11106161B2 | Intermediate transfer members for use with indirect printing systems and ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11106161B2/en>
- US10837947B2 | Modified resins and uses thereof | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10837947B2/en>
- US10844663B2 | Fiber reinforced elastomeric stator | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10844663B2/en>
- US9656446B2 | Articles and methods of making the same | Not disclosed | US | 2017 | <https://patents.google.com/patent/US9656446B2/en>
- US20180022951A1 | Electroconductive silver paste | Not disclosed | US | 2018 | <https://patents.google.com/patent/US20180022951A1/en>
- EP1921511B1 | Toner supply roller | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP1921511B1/en>